

Session 5 — Entrepreneurial Finance & Venture Capital

VC Term Sheets · Option Pool Mechanics · Liquidation Preferences · Fast Ion Battery | April 28, 2026

Section 1 Framing — The Big Questions

Before diving into term sheet mechanics, the session re-anchored on the two questions that drive all financing decisions — relevant to entrepreneurs and investors alike:

1. **How much external finance does this business need?** — Driven by the business model: cash curve, burn rate, fume date, and maximum financing need.
2. **How should I structure my financing?** — How much to raise in this round, what share to give up (valuation), which deal terms to accept, and who to take the money from.

Course arc so far: Sessions 1–2 built business model and cash flow intuition. Sessions 3–4 introduced valuation (DCF bottom-up and VC top-down). Session 5 goes deep on term sheet mechanics before returning to round-sizing and investor selection in later sessions.

Section 2 Goal of Studying Term Sheets

This is a finance class, not a law class. Lawyers focus on whether each term is legally watertight. Your job is to make a business decision — risk-adjusted return, cash allocation, and control allocation. Develop intuition around four questions:

- **Who gets paid, what, and when?** — Across all states of the world, not just the most likely outcome.
- **How is risk, cash, and control allocated?** — VC contracts deliberately separate cash flow rights from control rights. You may own the majority of the economics but cede operational control to preferred stockholders.
- **What are the longer-term consequences?** — Terms shape incentives going forward and determine who will (or will not) be attracted to invest in future rounds.
- **What is negotiable?** — Some terms are industry-standard; others are live negotiating points that meaningfully shift economic outcomes.

Section 3 Negotiating Leverage

A common fear is that entrepreneurs have no leverage. The professor addressed this directly:

VCs also need you. A VC issues a term sheet to roughly 1 in 100 deals they see. If they've done all the due diligence and partnership approval to reach you, they genuinely want to do the deal. If a change you request makes you more motivated and doesn't significantly hurt them, they will often agree.

Sequencing is everything. Don't negotiate exclusively with one investor until you have a term sheet and no other options. Instead:

Approach 30 potential investors simultaneously.

Advance conversations in parallel, keeping them roughly in sync.

When you receive a first term sheet (with its exploding deadline), go back to the others: 'I have a term sheet — are you still interested?' The dynamic shifts immediately.

Once others hear you have a competing term sheet, urgency and flexibility change dramatically — 'I'm your friend, take your time, what can I do to show I'm your value-added partner?'

Never show your cards early. Don't signal which terms you want to change before receiving the term sheet. If you ask for a post-financing option pool before the term sheet is written, the VC will simply lower the headline valuation to compensate — same economic outcome for them. The key saying: '{i('If you set the price and I set the terms, I can always win.')} Negotiate price and terms simultaneously.

Attitude matters. The best entrepreneurs treat fundraising as offering investors the chance to participate in a great opportunity — not begging for money. That posture changes the dynamic.

Section 4 The Cap Table — Baseline (Exhibit 1)

The session revisited the Exhibit 1 cap table from the prior session. Deal terms: \$2M raised, \$1/share, \$8M post-money, with an employee option pool of **12.5%** of fully-diluted post-money capitalisation created **concurrently** with the round.

Founders 62.5% ESOP 12.5% VCs 25.0%			
Pre-money \$6,000,000	Raised \$2,000,000	Post-money \$8,000,000	Share price \$1.00
Shareholder	Shares	Ownership	\$ Value
Founders (entrepreneur)	5,000,000	62.5%	\$5,000,000
ESOP (option pool)	1,000,000	12.5%	\$1,000,000
VCs (Delta + Omega)	2,000,000	25.0%	\$2,000,000
Total (post-money)	8,000,000	100.0%	\$8,000,000

Cross-check: $\text{pre-money} \div \text{pre-money shares} = \text{investment} \div \text{VC shares} = \text{post-money} \div \text{total shares} = \1.00 . If these disagree, you've made an error.

How to solve any cap table: Some facts are known, the rest follow by arithmetic. Here: VCs invested \$2M at \$8M post-money → they own 25% → 2M shares at \$1. Option pool = 12.5% of 8M = 1M shares. Founders get the rest: 5M shares / 62.5%.

Section 5 Option Pool Mechanics — Why Timing Is Everything

Three scenarios with the same \$8M post-money valuation and same \$2M investment show how the timing of the option pool creation shifts dilution — and therefore wealth — between founders and VCs.

Scenario A — No option pool

VCs still own 25% (25% of \$8M = \$2M). Founders own 75%. Since founders have 5M shares = 75%, total = 6,666,667 shares. VC gets 1,666,667 shares. Share price = \$8M divided by 6,666,667 = **\$1.20**.

Founders 75.0% VCs 25.0%			
Pre-money \$6,000,000	Share price \$1.20	Founder % 75.0%	VC % 25.0%
Shareholder	Shares	Ownership	\$ Value
Founders	5,000,000	75.0%	\$6,000,000
VCs	1,666,667	25.0%	\$2,000,000
Total	6,666,667	100.0%	\$8,000,000

No pool → higher share price (\$1.20 vs \$1.00). VC gets fewer shares because each costs more. Founders bear no option pool dilution.

Scenario B — Option pool created concurrently (Exhibit 1 — the standard VC approach)

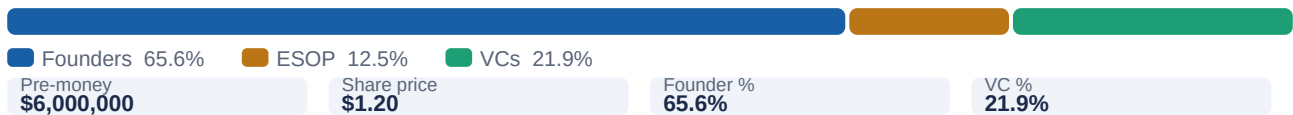
Option pool is created as part of the financing at the same pre-money step. The pool dilutes only the founder because it is carved out before the VC's price is set. The VC pays \$1/share and owns exactly 25%. All dilution from the pool falls on the founder.

Founders 62.5% ESOP 12.5% VCs 25.0%			
Pre-money \$6,000,000	Share price \$1.00	Founder % 62.5%	VC % 25.0%
Shareholder	Shares	Ownership	\$ Value
Founders	5,000,000	62.5%	\$5,000,000
ESOP	1,000,000	12.5%	\$1,000,000
VCs	2,000,000	25.0%	\$2,000,000
Total	8,000,000	100.0%	\$8,000,000

Standard VC structure. Founder drops from 75% → 62.5%. VC unchanged at 25%. The option pool diluted only the founder.

Scenario C — Option pool created after financing (entrepreneur-preferred)

Start from Scenario A (no pool). After the round closes, create a 12.5% ESOP. New shares dilute everyone on the cap table proportionally — both founders and VCs. If 6,666,667 is now 87.5%, the 12.5% pool requires 952,381 new shares. Total = 7,619,048.



Shareholder	Shares	Ownership	\$ Value
Founders	5,000,000	65.6%	\$5,250,000
ESOP	952,381	12.5%	\$1,000,000
VCs	1,666,667	21.9%	\$1,750,000
Total	7,619,048	100.0%	\$8,000,000

Founders: 65.6% vs 62.5% in Scenario B. VCs: 21.9% vs 25.0%. Post-financing pool dilutes both proportionally — this is why VCs insist on pre-financing pool creation.

The VC 'trick' — and how to counter it

- VC quotes a post-money valuation (sounds like a good headline). Then includes a large concurrent option pool (15–20%) within that post-money.
- This reduces the effective pre-money for founders — it becomes your shares + the pool, not just your shares.
- You cannot simply demand a post-financing pool as a tactic — VCs will just give you a lower headline valuation to compensate.
- The right move: wait for the term sheet, then negotiate the pool size downward. A smaller pre-financing pool = higher effective pre-money for founders.
- Key saying: *If you set the price and I set the terms, I can always win.* You must negotiate both simultaneously.

Section 5b What Happens at Acquisition — Unexercised Options

The cap table shows a 'fully diluted as-converted' picture — it assumes all authorised options have been granted, are in the money, and exercised. Reality is different. Options are the right, but not the obligation, to buy shares. If unexercised at acquisition, they simply vanish — no shares are created.

Example — acquisition with unexercised Exhibit 1 option pool:

VC converts 2M preferred → 2M common (1:1). Founder has 5M common. The 1M option pool was authorised but never exercised — it disappears.

Shareholder	Shares at acquisition	Actual %	Cap table %
Founders	5,000,000	71.4%	62.5%
VCs (converted)	2,000,000	28.6%	25.0%
Option pool (unexercised)	0	—	12.5%
Total actual	7,000,000	100.0%	100.0%

Cap table showed founders 62.5% / VCs 25%. Actual at acquisition: founders 71.4% / VCs 28.6%. Unexercised options don't dilute anyone.

Key implication for option pool timing: If the pool was created pre-financing (Scenario B) and the company is acquired before those options are exercised, the founder loses out compared to a world where the pool was created post-financing — because the pre-financing pool lowered the share price and thus diluted the founder even though no employee ever benefited.

Down-round options: Companies can go public below the valuation at which employee options were granted. Those options are underwater (strike price > market price) and will not be exercised. This happened to many cleantech companies post-2011.

Section 6 Liquidation Preferences — What Founders Actually Receive

Using Exhibit 1 (founders 62.5%, ESOP 12.5%, VCs 25%), the class computed actual cash payouts in a \$20M acquisition. All options assumed fully granted and exercised at \$0.10/share. Acquisition share price = \$20M ÷ 8M shares = \$2.50.

Why \$0.10 exercise price? Options must be granted at fair market value of common stock. Common legitimately trades at a discount to preferred — it lacks liquidation preference, control rights, and anti-dilution protection. At early stage this discount can be ~90%. The gap converges at IPO when preferred converts to common at the same price.

Liquidation preference scenarios — \$20M acquisition

Scenario	VC payout	Founder payout	Options payout
1× convertible preferred (converts to common)	\$5.0M $25\% \times \$20M$	\$12.5M $62.5\% \times \$20M$	\$2.4M $1M \times (\$2.50 - \$0.10)$
1× participating convertible preferred (liq pref + pro-rata)	\$6.5M $\$2M + 25\% \times \$18M$	\$11.3M $62.5\% \times \$18M$	~\$2.2M $12.5\% \times \$18M - \text{exercise}$
2× participating convertible preferred (double liq pref + pro-rata)	\$8.0M $\$4M + 25\% \times \$16M$	\$10.0M $62.5\% \times \$16M$	~\$1.9M $12.5\% \times \$16M - \text{exercise}$

Post-money valuation is \$8M in all three cases. Only the term structure changes. Founder payout falls \$12.5M → \$10.0M as terms escalate.

Scenario 1 — Convertible preferred, 1× liquidation preference:

At a \$20M acquisition, the VC can: (a) take their 1× preference (\$2M) and stop, or (b) convert to common and participate pro-rata ($25\% \times \$20M = \$5M$). They choose (b) since $\$5M > \$2M$. Founders: $62.5\% \times \$20M = \$12.5M$. Options: $1M \times (\$2.50 - \$0.10) = \$2.4M$.

Scenario 2 — Participating convertible preferred, 1× liquidation preference:

VC takes \$2M preference first (the 'double-dip'), then participates in remaining \$18M at 25% → \$4.5M. Total to VC = \$6.5M. Founders get 62.5% of \$18M = \$11.3M. The double-dip costs founders \$1.2M vs Scenario 1.

Scenario 3 — Participating convertible preferred, 2× liquidation preference:

VC takes \$4M first (2× their \$2M investment), then participates in \$16M at 25% → \$8M total. Founders: $62.5\% \times \$16M = \$10.0M$. Each escalation in preference transfers value from founders to VCs.

Key takeaways on liquidation preferences

- The headline post-money valuation is identical in all three scenarios.
- Participating preferred is far more valuable to the VC than non-participating, especially at moderate exits.
- A higher liquidation multiple (2×, 3×) further concentrates returns to the VC.
- These terms are negotiable — push for non-participating preferred or at minimum a cap on participation.

Section 7 Fast Ion Battery — Case Study (HBS 9-815-025)

Background

Fast Ion Battery was founded in May 2008 by an MIT post-doctoral researcher. The technology addressed a fundamental storage gap — as Bill Gates noted: 'all the batteries on Earth could currently store less than 10 minutes of the world's electricity needs.' John Davidson (WSC) and Donna Lerner (Bluelock) — both HBS Class of 2005, personal friends — co-led the Series A alongside Franconia Ventures.

The financing structure — tranching Series A

Davidson and co-investors tranching the \$10M Series A to manage technology risk, with the second tranche conditional on meeting specific milestones:

Round	Date	Amount	Valuation	Key milestones / status
Series A1	May 2008	\$4M	\$5M pre / \$9M post	Prove tech reproducibility, IP portfolio, define customer segments
ARPA-E grant	Dec 2009	\$2M non-dilutive	—	Competitive grant — provided validation and non-dilutive capital
Series A2	Dec 2009	\$6M	\$13.5M pre / \$19.5M post (1.5× step-up)	Scale-up, business model, firm orders from partners, complete team
Bridge (proposed)	Dec 2011	\$5M	Flat (same price as A2)	Hire new CEO, secure missed A2 milestones
Series B (projected)	Dec 2012	\$25M	TBD	Pilot manufacturing, begin revenue
Series C (projected)	Jun 2014	\$35M	TBD	Full manufacturing, \$100M revenue capacity

Total projected equity ~\$75M. Target exit: \$500M+ (EV batteries + grid-scale storage).

Actual progress vs milestones

Round	Target	Actual
Series A1	Reproduce tech outside lab IP strategy Define customer segments	Won ARPA-E grant ✓ Secured IP portfolio ✓ Customer segments not defined ✗
Series A2	Scale-up demonstrated Firm orders from partners Build engineering & sales team Hire strong CEO	Scale-up potential shown ✓ Capital-light manufacturing model defined ✓ No hard orders/purchase orders ✗ CEO being replaced ✗

Davidson: 'It was not quite right and not quite wrong either. The technology continued to have promise... we were heartened by the ARPA-E validation.'

Davidson's projected cap table (Exhibit 2) — the bull case

Davidson's investment memo projected ownership evolution from A1 through a potential Series C exit. WSC projected to invest \$11.25M and own ~18.8% at exit — implying a 5.8× return at 45% IRR (assuming a \$350M exit valuation):

Shareholder	A1 Shares	A1 %	A2 Shares	A2 %	Series B Shares	B %
Common (founders + pool)	5,000,000	55.6%	5,000,000	38.5%	5,000,000	25.6%
WSC (A1)	1,500,000	16.7%	1,500,000	11.5%	1,500,000	7.7%
Franconia (A1)	1,500,000	16.7%	1,500,000	11.5%	1,500,000	7.7%
Bluelock (A1)	1,000,000	11.1%	1,000,000	7.7%	1,000,000	5.1%
WSC (A2)	—	—	1,500,000	11.5%	1,500,000	7.7%
Franconia (A2)	—	—	1,500,000	11.5%	1,500,000	7.7%
Bluelock (A2)	—	—	1,000,000	7.7%	1,000,000	5.1%
Series B investors	—	—	—	—	3,900,000	20.0%
Total	9,000,000	100%	13,000,000	100%	19,500,000	100%

Exhibit 2 from Davidson's original investment memo to WSC partners.

Section 8 The Cleantech Problem — Investment Climate in 2011

By December 2011, the cleantech investment thesis had fundamentally broken down. The Solyndra bankruptcy was the symbol of broader failure. LPs had grown wary; most VCs were shifting capital to software, SaaS, and consumer internet — sectors where the 'cost of learning' had collapsed due to cloud computing and open source.

Why cleantech is structurally mismatched with VC (Exhibit 6):

Cleantech follows a capital-intensive development cycle with two distinct 'valleys of death' before commercialisation, spanning 10–40+ years from basic research to mass deployment. This timeline is fundamentally incompatible with typical VC fund life (10 years).

In contrast, software/SaaS/consumer internet companies can test market hypotheses cheaply and iterate quickly — the cost of learning has collapsed. VC is well-suited to this profile.

VC deal flow data (Exhibit 7):

Sector	% of deals 2005	% of deals 2011	2012 forecast
Software & services	42%	45%	Growing — 60%+ of VCs predict increases
Computer hardware & equipment	24%	11%	Declining
Biotech & medical devices	21%	21%	58% of VCs predict decreases
Energy & industrials (cleantech)	5%	8%	55% of VCs predict decreases

Source: National Venture Capital Association / MoneyTree Q4 2011.

A123 Systems as a cautionary tale (Exhibit 8):

A123 Systems (battery company, comparable to Fast Ion) raised \$351M across 7 rounds before its \$428M IPO in 2009. By November 2011, the stock had fallen 84% from IPO price.

Investor	Pre-IPO %	Post-IPO %	Observation
Founders	5.6%	4.1%	Severe dilution from 7 rounds of capital
Options pool	15.5%	11.2%	—
Northbridge	10.6%	7.7%	Lead VC
GE	9.9%	7.3%	Strategic investor
Other investors	34.6%	25.0%	—
IPO new shares	—	27.4%	\$428M raised at \$13.50/share

Total pre-IPO capital raised: \$351M across 7 rounds. Key lesson: cleantech requires massive, repeated capital raises with modest step-ups, leading to extreme dilution for early investors.

IPO performance — cleantech vs software/internet (Exhibit 9):

Company	Sector	IPO year	IPO price	Nov 2011 price	% change
Ascent Solar	Photovoltaics	2006	\$5.50	\$7.10	+29%
Canadian Solar	Photovoltaics	2006	\$15.00	\$3.36	−78%
A123 Systems	Batteries	2009	\$13.50	\$2.10	−84%
Generac Holdings	Generators	2010	\$13.00	\$24.64	+90%
SemiLEDs	LEDs	2010	\$17.00	\$3.60	−79%
Baidu.com	Internet search	2005	\$27.00	\$146.41	+442%
LinkedIn	Social network	2011	\$32.00	\$87.50	+173%
Zillow	Online real estate	2011	\$18.00	\$32.63	+81%
Pandora Media	Music streaming	2011	\$16.00	\$9.85	−38%

CleanTech IPOs showed mixed/negative returns by Nov 2011. Software and internet IPOs generally outperformed dramatically.

Section 9 The Bridge Round — Terms and Investor Dynamics

By December 2011 Fast Ion was weeks away from missing payroll. Davidson had emailed Lerner (Bluelock) and Mehra (Franconia) to discuss a \$5M bridge. Lerner's email response: concern about CEO search pace, investment committee squeamishness, and lower valuation step-ups compared to software investments.

Then came the call: 'I'm sorry, John. The cleantech thesis has not panned out for us and we're under pressure to limit our exposure. We cannot continue supporting Fast Ion.'

Proposed bridge terms (Exhibit 4):

Term	Detail
Amount	Minimum \$5M
Pro-rata split	WSC: \$1,875,000 Franconia: \$1,875,000 Bluelock: \$1,250,000
Security / Price	Series A2 Preferred — flat round (same price as A2)
Terms	Identical to existing A2 Preferred Stock
Oversubscription incentive	If investor buys > pro-rata: receives common stock warrants = oversubscription ÷ A2 price. Exercise price \$0.50/share, 10-year life
Pay-to-play	If investor does NOT participate to full pro-rata: ALL existing preferred stock converts to common at 10:1 ratio (10 preferred → 1 common). A severe penalty.

Pay-to-play is the enforcement mechanism — it creates a powerful incentive to participate. If Bluelock declines, their entire preferred position is wiped down 10:1.

Why the two firms faced different decisions:

	Ware Street Capital (WSC)	Bluelock Ventures
Fund stage	Brand new \$100M fund	5 years into a \$500M fund
Fund performance	<5% net IRR (only 1 other deal)	Already returned >\$1B
Partner track record	Davidson: still working on his first home run	Lerner: already had 2 home runs
Max deal size	\$10–12M per deal	Up to \$25M per deal
Incentive	Fast Ion is Davidson's potential home run — high motivation to save it	Less pressure; other exits; cleantech seen as a drag on fund performance

Structurally different incentives explain the divergence — not just different views on Fast Ion's prospects.

Davidson's dilemma — bull vs bear case:

Bull case	Bear case
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Two excellent CEO candidates in final stages. Technology gaining traction in EV batteries. Potential to dominate EV and ultimately grid-scale storage. Davidson's target: \$500M exit on \$75M total investment (~6.7×). Bridge buys 12–15 months of runway to reach that inflection.

Company already consumed \$10M without meeting key milestones. Hostile cleantech investment climate. Without Bluelock's \$1.25M, WSC must cover their share too. Would later-stage investors put in \$25M+ for Series B in this environment? Risk of throwing good money after bad.

Davidson's reflection: 'There are a hundred different ways I can lose all my money on an investment, but as a venture capitalist investing in risky startups that is par-for-the-course... my job is to see if there is a way to make Fast Ion such a company and to own a large enough share of it to make the risk worthwhile.'

Section 10 The 'Bad' Term Sheet — Newmatco Angel Deal

The breakout session analysed a real term sheet given by the HBS Alumni Angel Association to a former HBS student, who shared it with the professor for teaching purposes. First, the class reconstructed the cap table:



Newmatco cap table at closing

Shareholder	Shares	Price/share	\$ Invested	Ownership %
Founders	320,000	\$2.75	—	58.67%
ESOP pool	80,000	\$2.75	\$220,000	14.67%
Angels	145,455	\$2.75	\$400,000	26.67%
Total (post-money)	545,455	—	\$400,000	100.0%

Pre-money: \$1.1M | Post-money: \$1.5M | Price: \$2.75/share. Check: pre-money shares (320K + 80K = 400K) × \$2.75 = \$1.1M ✓

Plus: 500,000 warrants at \$5.00 exercise price, 7-year life

Currently out of the money (\$5.00 exercise vs \$2.75 current price) but with a 7-year life, they have real option value. Any future investor must assume these will eventually be exercised and include them in the fully-diluted cap table:

Fully-diluted cap table (including warrants)

Shareholder	Shares (basic)	Warrants	Shares (fully diluted)	% fully diluted
Founders	320,000	—	320,000	30.62%
ESOP pool	80,000	—	80,000	7.66%
Angels	145,455	500,000	645,455	61.73%
Total	545,455	500,000	1,045,455	100.0%

Although warrants are currently out of the money, future investors will assume they will be exercised. On this basis angels effectively control 61.7% of the company. The post-money valuation is still quoted as \$1.5M — post-money valuation is an incomplete and often misleading headline figure.

Why warrants are 'unbelievably attractive' for investors:

Warrants are a right, not an obligation. No cash is put down upfront. If the company does well and the share price exceeds \$5.00, the angels simply exercise — paying \$5 per share — and pocket the difference between the exercise price and market price. This is an extraordinarily capital-light way to gain leveraged upside with no downside risk. The warrants are free — given as part of the term sheet with no additional payment.

The problematic terms — full breakdown:

Full ratchet anti-dilution	The most aggressive protection. If any future share is issued at a price below \$2.75, the angels' conversion price resets to that lower price — giving them far more shares. In a capital-intensive material-science venture, this makes future investors extremely reluctant to do down-rounds, potentially killing all future financing.
12% cumulative compounding dividends	Not just cumulative (unpaid dividends accrue) but compounding — the interest itself earns interest. At 12% p.a. compounded, this grows rapidly. These dividends take priority at any liquidation event. Compare: non-cumulative 8% (as in Fast Ion A1) is reasonable; 12% compounding is egregious.
2× participating convertible preferred	Angels take 2× their \$400K investment (\$800K preference) first, then participate pro-rata alongside common. At a modest exit this could consume all proceeds before founders see anything.
2× pro-rata participation right in future rounds	Angels own 26.7% → they can take up to 53.4% of any future round. This effectively shuts out new investors who cannot get meaningful ownership. Makes the company un-fundable in subsequent rounds.
Angel-stacked board	Board composition skewed to angel-appointed directors, giving effective control over major decisions despite owning <27% economically.
Egregious information rights	Any individual angel can request facility tours, bring guests, and demand reporting at unusually high frequencies. Operationally disruptive and highly non-standard.
500,000 warrants at \$5.00	While currently out of the money, these give the angels the right to own >61% on a fully-diluted basis. Future investors must include them in cap table projections, making the angels appear to control the majority of the company.

Why might angels offer a term sheet like this?

1. **Naivety:** Angels without deep VC experience don't always understand downstream consequences.
2. **Greed:** Simply extracting as much as possible.
3. **Rational fear of dilution (most charitable view):** Material-science companies are capital-intensive. Angels know they cannot follow through multiple large VC rounds — they will be diluted severely. They're trying to protect their economics given that dilution trajectory. The problem: doing so in this way kills the company by making it un-fundable, which is ultimately self-defeating for the very investment they're protecting.

The professor's verdict

- Even if this is your only term sheet, you may be better off refusing it.
- This term sheet is a 'poison pill' — taking it may render the company un-fundable in all future rounds.
- It is better to walk away and seek alternative funding than accept terms that destroy future financing options.
- This was a **real term sheet** given by the Harvard Business School Alumni Angel Association to an actual HBS graduate.

Section 11 Series A Term Sheet — Fast Ion (Exhibit 1 of the case)

For contrast with the Newmatco bad term sheet, the Fast Ion Series A terms (Exhibit 1 of the case) represent a more standard, reasonable set of VC terms:

Term	What it says	Assessment
Security	Series A1 and A2 Preferred Stock	Standard
Dividend	8% non-cumulative, payable if declared by Board	Reasonable — non-cumulative means unpaid dividends do not accrue
Liquidation preference	1× preference, non-participating. Remaining assets to common.	Reasonable — non-participating; founder gets upside above 1× threshold
Conversion	Any time, 1 preferred → 1 common	Standard
Anti-dilution	Broad-based weighted average	Reasonable — less aggressive than full ratchet
Board	3 seats: CEO, Davidson (investor rep), 1 independent	Balanced — founder/investor/independent split
ESOP vesting	4-year with 12-month cliff	Standard
Founder vesting	25% at 1 year, remainder monthly over 36 months	Standard
Pay-to-play	Not included in Series A terms	Note: proposed in bridge round with severe 10:1 conversion penalty
Drag-along	After 60 months, investors can require founders to sell at same price	Standard protective right
ROFR / Co-sale	Company and investors have right of first refusal on founder share sales	Standard protective right

Fast Ion Series A terms are broadly reasonable — the key differentiators vs Newmatco: non-cumulative dividend, non-participating preferred, broad-based weighted-average anti-dilution, and balanced board.

Section 12 Key Takeaways

1

Post-money valuation is a headline, not the whole story. The same \$8M post-money can imply radically different economic outcomes depending on liquidation preference structure, option pool size, and warrant overhang. Never evaluate a deal on valuation alone.

2

Option pool timing is a wealth transfer mechanism. Creating the pool concurrently with (or before) financing dilutes only the founder. Post-financing pool creation dilutes everyone proportionally. VCs systematically prefer the pre-financing pool. Push back by negotiating the pool size downward — a smaller pool = higher effective pre-money for founders.

3

Unexercised options vanish on acquisition. The cap table is a fully-diluted best-case picture. In practice, if the company is acquired before options are exercised, those shares do not exist — actual ownership percentages can be significantly better or worse than the cap table implies.

4	Liquidation preferences compound. 1× non-participating < 1× participating < 2× participating. Each escalation transfers value from founders to VCs, particularly at moderate exits. These are negotiable — push for non-participating preferred at minimum.
5	Warrants are extraordinarily valuable. They provide leveraged upside with no downside risk and no upfront cost. Future investors must include them in fully-diluted calculations. A warrant overhang can fundamentally change the cap table picture even when currently out of the money.
6	Negotiating leverage comes from sequencing. Run parallel processes with 30 investors. Never show your cards before receiving the term sheet. The moment you have a competing term sheet, the entire dynamic shifts in your favour.
7	Cleantech is structurally mismatched with the VC model. Multiple capital-intensive valleys of death, 10–40 year timelines, and modest step-ups between rounds create extreme dilution for early investors. Fast Ion and A123 Systems illustrate why most VCs exited cleantech after 2008–2011.
8	A bad term sheet may be worse than no deal at all. Full ratchet anti-dilution, cumulative compounding dividends, 2× participating preferred, and aggressive pro-rata rights can collectively make a company un-fundable. Walking away may be the right answer even if it is the only offer on the table.